

Lecture 14: Distribution Shift and Double Robustness

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Last Class Recap: Covariate Shift

Setup. Let P be the source distribution and Q be the target distribution, both over (x, y) .

Covariate shift assumption:

$$P(y \mid x) = Q(y \mid x), \quad \text{but} \quad P_x \neq Q_x.$$

Ideal setting: If $\text{supp}(Q_x) \subseteq \text{supp}(P_x)$, we can use *weighted ERM*.

Solutions

(1) **Reweighting.** Match aggregate statistics $\Phi(x)$:

$$\min_{\pi} \text{KL}(\pi \parallel P_{\text{ref}}) \quad \text{such that} \quad \mathbb{E}_{\pi}[\Phi(x)] = t.$$

(2) **Move the points (Optimal Transport).** Transport source points $x_i \rightarrow \tilde{x}_i$ (in the Q_T sense):

$$\min_{\pi} \sum_{i,j} C_{ij} \Pi_{ij} \quad \text{s.t.} \quad \Pi \mathbf{1} = p, \quad \Pi^{\top} \mathbf{1} = q.$$

Population Loss Under Covariate Shift

The population loss we minimize is:

$$\min_{h \in \mathcal{H}} \mathbb{E}_{(x,y) \sim P} [\omega(x) \text{loss}(h(x), y)],$$

where the *importance weight* is

$$\omega(x) = \frac{Q(x)}{P(x)}.$$

Issues:

1. Hard to estimate the ratio of densities.
2. High variance.

Entropic Optimal Transport

To reduce variance, add an entropy regularizer:

$$\min_{\Pi} \sum_{i,j=1}^n C_{ij} \Pi_{ij} - \varepsilon H(\Pi) \approx \varepsilon \min_{\Pi} \text{KL}(\Pi \| K), \quad \text{s.t.} \quad \Pi \mathbf{1} = p, \Pi^\top \mathbf{1} = q,$$

where $K_{ij} = \exp(-C_{ij}/\varepsilon)$ is the Gibbs (Sinkhorn) kernel. Solved efficiently by Sinkhorn matrix scaling [Chewi, Neals, Weed, Rigollet].

Here $C_{ij} = \|x_i - x'_j\|_2$, with $x_1, \dots, x_n$: the source points and $x'_1, \dots, x'_n$: the target points.

Causal Inference and Treatment Effects

Setup.

- Treatment $T \in \{0, 1\}$, covariates/features x , outcome $y \in \mathbb{R}$.
- Samples from joint distribution (X, T, Y) .

Average Treatment Effect (ATE):

$$\theta = \mathbb{E}[y(1)] - \mathbb{E}[y(0)],$$

where $\theta_1 = \mathbb{E}[y(1)]$ (treated potential outcome) and $\theta_0 = \mathbb{E}[y(0)]$ (control).

Given samples $(x_1, T_1, y_1), \dots, (x_n, T_n, y_n)$, we wish to estimate θ .

Ideal case: Randomized Controlled Trial (RCT). If $T \perp X$, then

$$\mathbb{E}[y \mid T = 1] = \mathbb{E}[y(1) \mid T = 1] = \mathbb{E}_x \mathbb{E}[y(1) \mid x] = \mathbb{E}[y(1)].$$

Key assumption: No unmeasured confounding (“no planche effect”).

$$y(1) \perp T \mid x.$$

Treated units $\neq$ whole population, so

$$\mathbb{E}[y \mid T = 1] \neq \mathbb{E}[y(1)] := \theta_1,$$

because $T \not\perp x$ in general. This is a **covariate shift** problem: source distribution $\neq$ target distribution.

(1) Propensity score.

$$e^*(x) = \Pr[T = 1 \mid X = x].$$

This plays the role of importance weights:

$$\omega(x) = \frac{p_X(x)}{p_{X|T=1}(x)} = \frac{\Pr[T = 1]}{e^*(x)}.$$

(2) Treated-outcome regression.

$$m_1^*(x) = \mathbb{E}[y \mid X = x, T = 1].$$

Assumptions on (X, T, Y)

1. **No hidden confounder:** outcome and treatment are conditionally independent given x ,

$$y(1) \perp T \mid x.$$

2. **Overlap:**

$$e^*(x) = \Pr[T = 1 \mid X = x] > 0.$$

Key Claim

Claim. $m_1^*(x) = \mathbb{E}[y(1) \mid X = x]$, so

$$\theta_1 = \mathbb{E}_x[m_1^*(x)].$$

Proof.

$$m_1^*(x) = \mathbb{E}[y \mid X = x, T = 1] = \mathbb{E}[y(1) \mid X = x, T = 1] \stackrel{\text{no confounder}}{=} \mathbb{E}[y(1) \mid X = x].$$

Hence,

$$\theta_1 = \mathbb{E}[y(1)] = \mathbb{E}_x \mathbb{E}[y(1) \mid x] = \mathbb{E}_x[m_1^*(x)].$$

□

Estimators for the ATE

Idea 1: Regression Adjustment

Learn $\hat{m}_1$ from samples $\{(x_i, T_i, y_i) : i \in [n]\}$ with $T_i = 1$. Then estimate:

$$\hat{\theta}_{\text{reg}} = \frac{1}{n} \sum_{i=1}^n \hat{m}_1(x_i).$$

If $\hat{m}_1 = m_1^*$, then $\mathbb{E}[\hat{\theta}_{\text{reg}}] = \theta_1$. But $\hat{m}_1 \neq m_1^*$ in general.

Issue. The training distribution of x (only treated units) differs from the target distribution (whole population):

$$\begin{aligned} \mathbb{E}_x[\hat{m}_1(x) - m_1^*(x)] &\neq \mathbb{E}_{x|T=1}[\hat{m}_1(x) - m_1^*(x)], \\ \mathbb{E}_x[(\hat{m}_1(x) - m_1^*(x))^2] &\neq \mathbb{E}_{x|T=1}[(\hat{m}_1(x) - m_1^*(x))^2]. \end{aligned}$$

Idea 2: Inverse Propensity Weighting (IPW)

Use a black-box model $\hat{e}$ approximating e^* . Estimator:

$$\hat{\theta}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i y_i}{\hat{e}(x_i)}.$$

If $\hat{e} = e^*$:

$$\mathbb{E}[\hat{\theta}_{\text{IPW}}] = \mathbb{E}\left[\frac{Ty}{\hat{e}(x)}\right] = \mathbb{E}_x\left[\mathbb{E}[Ty \mid x] \cdot \frac{1}{\hat{e}(x)}\right] = \mathbb{E}_x\left[\frac{e^*(x)}{\hat{e}(x)} m_1^*(x)\right] = \theta_1.$$

But $\hat{e} \neq e^*$ in general, so the estimator may be biased.

Doubly Robust Estimator (AIPW)

The *Augmented Inverse Propensity Weighted* (AIPW) estimator combines both approaches:

$$\hat{\theta}_{\text{DR}} = \frac{1}{n} \sum_{i=1}^n \hat{m}_1(x_i) + \frac{1}{n} \sum_{i=1}^n \frac{T_i (y_i - \hat{m}_1(x_i))}{\hat{e}(x_i)}.$$

Double Robustness Theorem

$$\mathbb{E}[\hat{\theta}_{\text{DR}}] - \theta_1 = \mathbb{E}\left[(\hat{m}_1(x) - m_1^*(x)) \cdot \frac{e^*(x) - \hat{e}(x)}{\hat{e}(x)}\right].$$

Suppose $\hat{e}(x) \geq \eta > 0$. Then:

$$|\mathbb{E}[\hat{\theta}_{\text{DR}}] - \theta_1| \leq \frac{1}{\eta} \|\hat{m}_1 - m_1^*\|_{L_2} \|e^* - \hat{e}\|_{L_2}.$$

In particular, $\hat{\theta}_{\text{DR}}$ is *doubly robust*: it is consistent if *either* $\hat{m}_1 \rightarrow m_1^*$ *or* $\hat{e} \rightarrow e^*$ (but not necessarily both).

Proof Sketch

Proof. We compute $\mathbb{E}\left[\frac{T(y - \hat{m}_1(x))}{\hat{e}(x)}\right]$ by conditioning on x :

$$\mathbb{E}\left[\frac{T(y - \hat{m}_1(x))}{\hat{e}(x)}\right] = \mathbb{E}_x\left[\frac{e^*(x)}{\hat{e}(x)} \cdot \mathbb{E}[(y - \hat{m}_1(x)) \mid x, T = 1]\right] = \mathbb{E}_x\left[\frac{e^*(x)}{\hat{e}(x)} (m_1^*(x) - \hat{m}_1(x))\right].$$

Adding the regression term:

$$\begin{aligned} \mathbb{E}[\hat{\theta}_{\text{DR}}] - \theta_1 &= \mathbb{E}_x[\hat{m}_1(x)] + \mathbb{E}_x\left[\frac{e^*(x)}{\hat{e}(x)} (m_1^*(x) - \hat{m}_1(x))\right] - \mathbb{E}_x[m_1^*(x)] \\ &= \mathbb{E}_x\left[(\hat{m}_1(x) - m_1^*(x)) \left(1 - \frac{e^*(x)}{\hat{e}(x)}\right)\right] \\ &= \mathbb{E}_x\left[(\hat{m}_1(x) - m_1^*(x)) \cdot \frac{e^*(x) - \hat{e}(x)}{\hat{e}(x)}\right]. \end{aligned}$$

□

This product structure is an instance of **Neyman Orthogonality**: the bias of $\hat{\theta}_{\text{DR}}$ is a *product* of the errors in estimating m_1^* and e^* , so the estimator tolerates slow convergence in each nuisance component.